

inFORM: The Future of Touchable 3D Displays

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Introduction / Reason

3D displays have great potential

- show 3D information
- Touch digital information

If they can present temperature, touch and etc...

→ **Displays aren't all that different from the real thing**

1. History & Description

Conventional 3D displays

- Only focused on showing the shape

→ **inForm** focused on **touchable UI** and Dynamic Physical Affordances

1. History & Description

UI examples

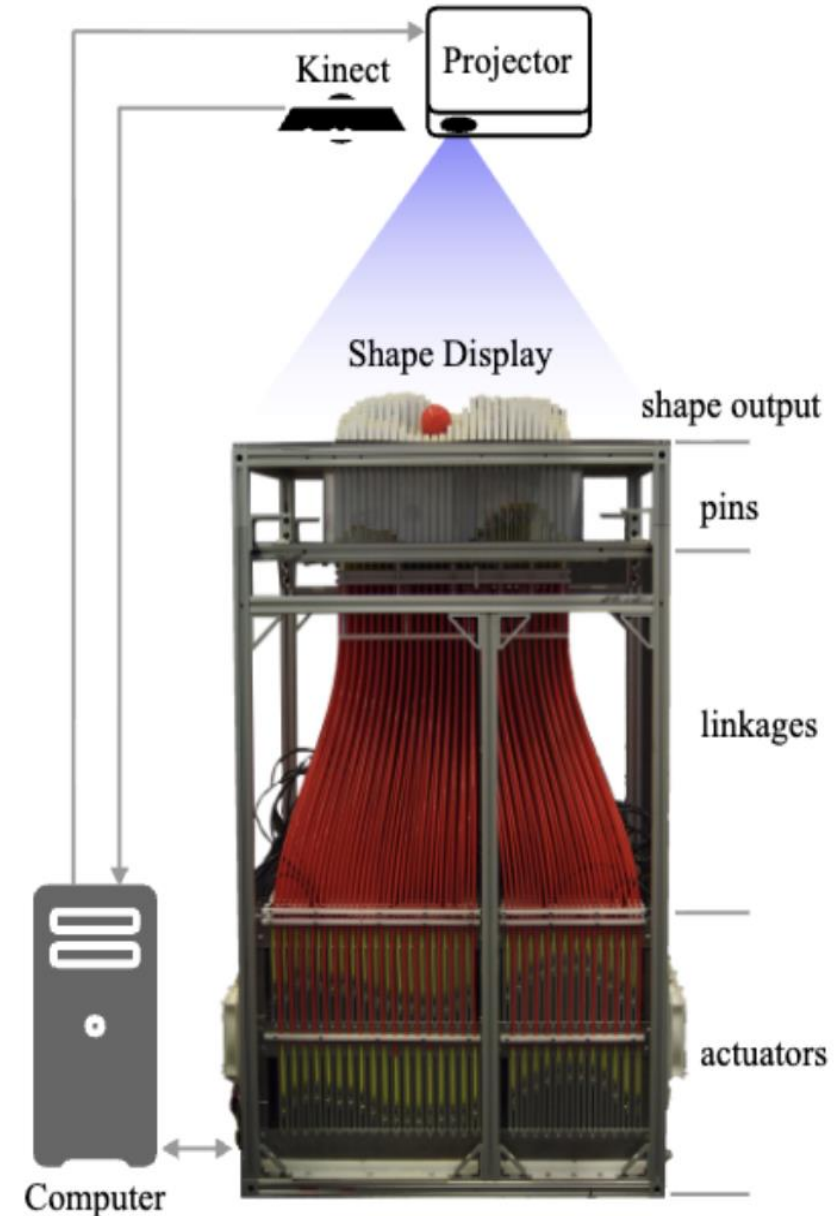
- GUI → Merit : Dynamic , Demerit : 2D
 - Tangible Interface → Merit : 3D , Demerit : Not Dynamic
- inForm : Dynamic and 3D

Procedure



Features

- Consisting of 30×30 pins and its size are $381 \times 381\text{mm}$
- Each pin extends up to 100 mm
- Each pin is moved by the actuator located below it
- Not only does it move the pin, but it also tells you where it is



Applications

1. Dynamic Physical UI

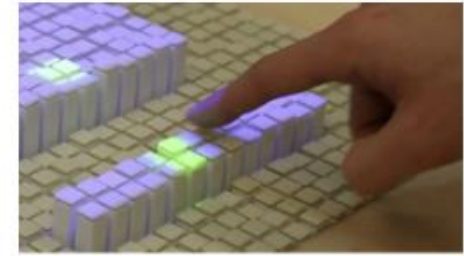
- UI can appear only when needed and

2. Interactive Surfaces

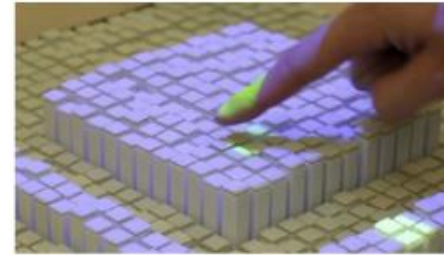
- Surfaces change shape depending on user actions.



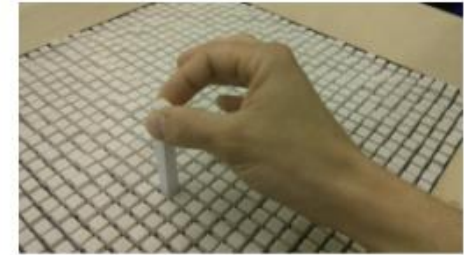
(a) Button



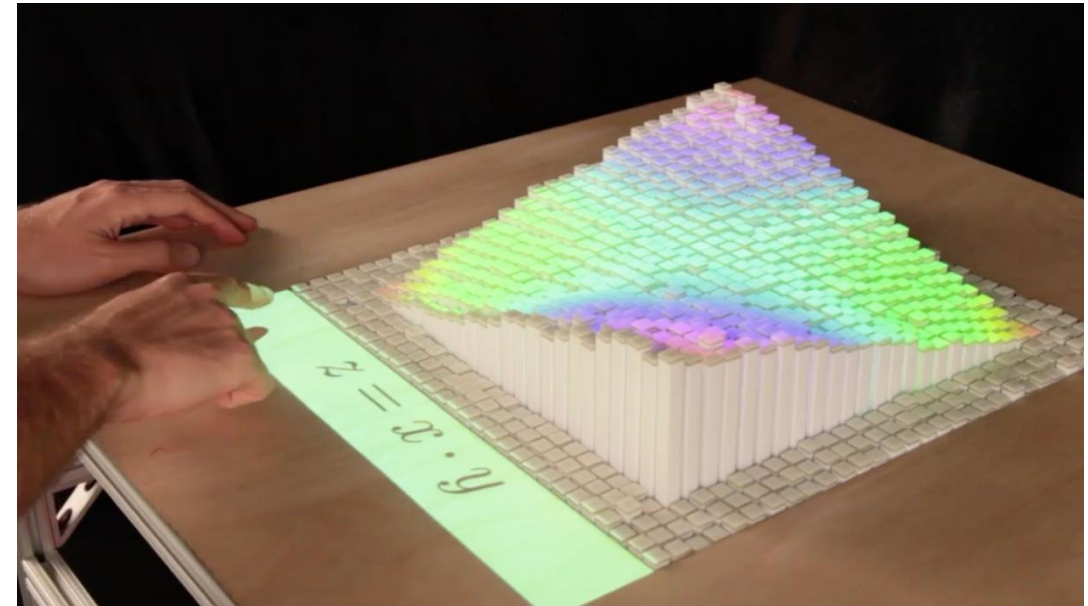
(b) 1D Touch Track



(c) 2D Touch Surface



(d) Handle



Applications

3. Dynamic Constraints

- Guide or restrict object movement.

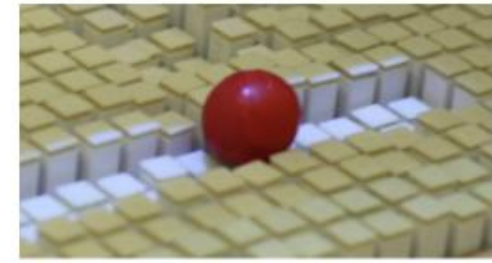
- inFORM connects digital information and the physical world.

4. Remote Physical Interaction

- Represent objects or movements from a distant location.



(a) Well



(b) Slot

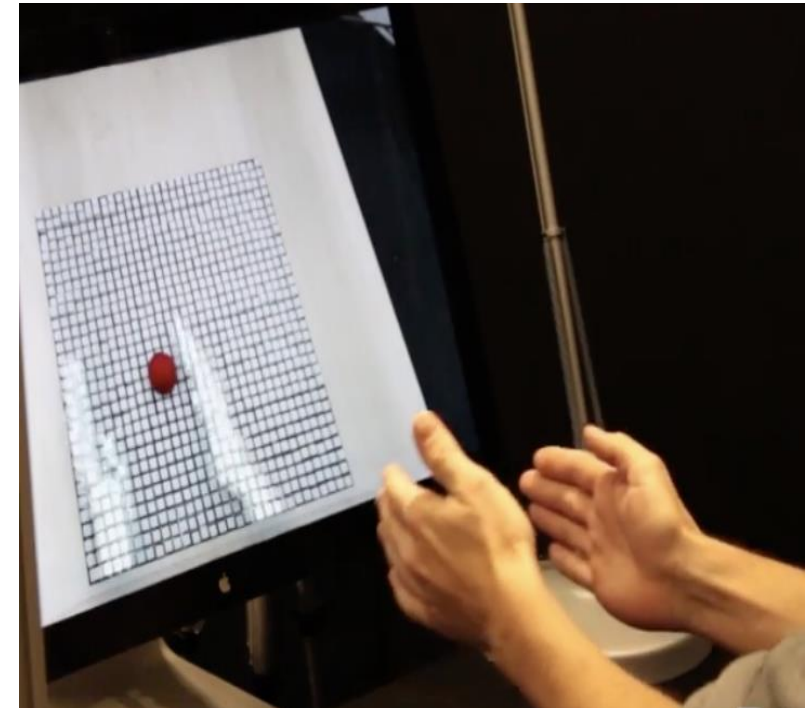
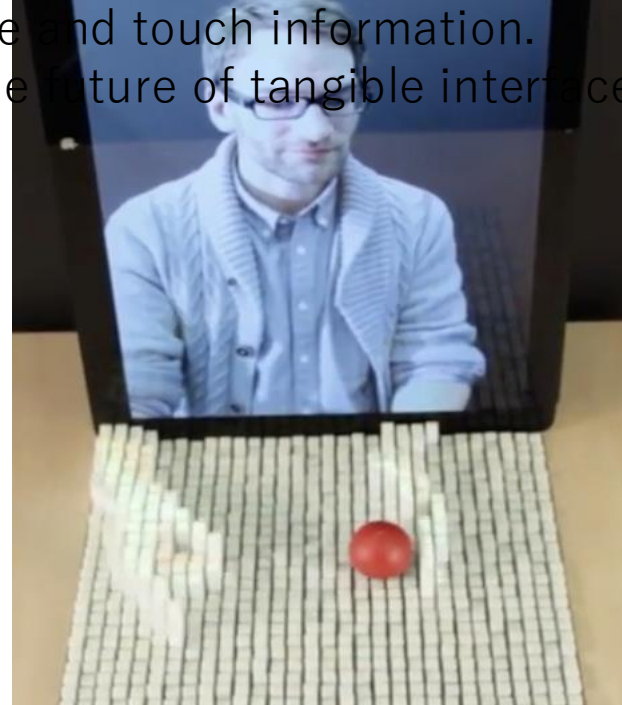


(c) Ramp



(d) Surface

- Users can both see and touch information.
- It demonstrates the future of tangible interfaces.



Impact on Society

Future Possibilities

- More intuitive human-computer interaction
- Better education and training systems
- Improved accessibility
- New ways of remote collaboration

Conclusion

- inFORM connects digital information and the physical world.
- Users can both see and touch information.
- It demonstrates the future of tangible interfaces.

Information source

A. inFORM: Dynamic Physical Affordances and Constraints through Shape and Object Actuation

B. UIST 2013

C. Page numbers: 417,423&424

D. URL: <https://dl.acm.org/doi/abs/10.1145/2501988.2502032>